

Uses of a Multimeter in Troubleshooting

no	Real-life scenario	how the multimeter helps
1	Pc power supply problem	A multimeter can measure the voltage from the power supply to check whether it is providing the expected output.
2	Laptop charger not working	A multimeter can check the charger's output voltage to help identify whether the charger is working properly.
3	Smartphone charging problem	A multimeter can be used to check voltage and continuity in the charging circuit to help find a faulty connection or component.

Conclusion

A multimeter is useful for checking electrical values such as voltage, resistance, and continuity. It can help identify whether a power or connection problem is causing a device to malfunction.

Task 2: POST Card Demonstration

I watched a POST card demonstration to understand how it helps in motherboard troubleshooting. The POST card displays hexadecimal codes while the BIOS checks different hardware components during startup. The code changes as different hardware tests are performed. If the system stops at a particular code, it can help identify the hardware area where the problem may be located.

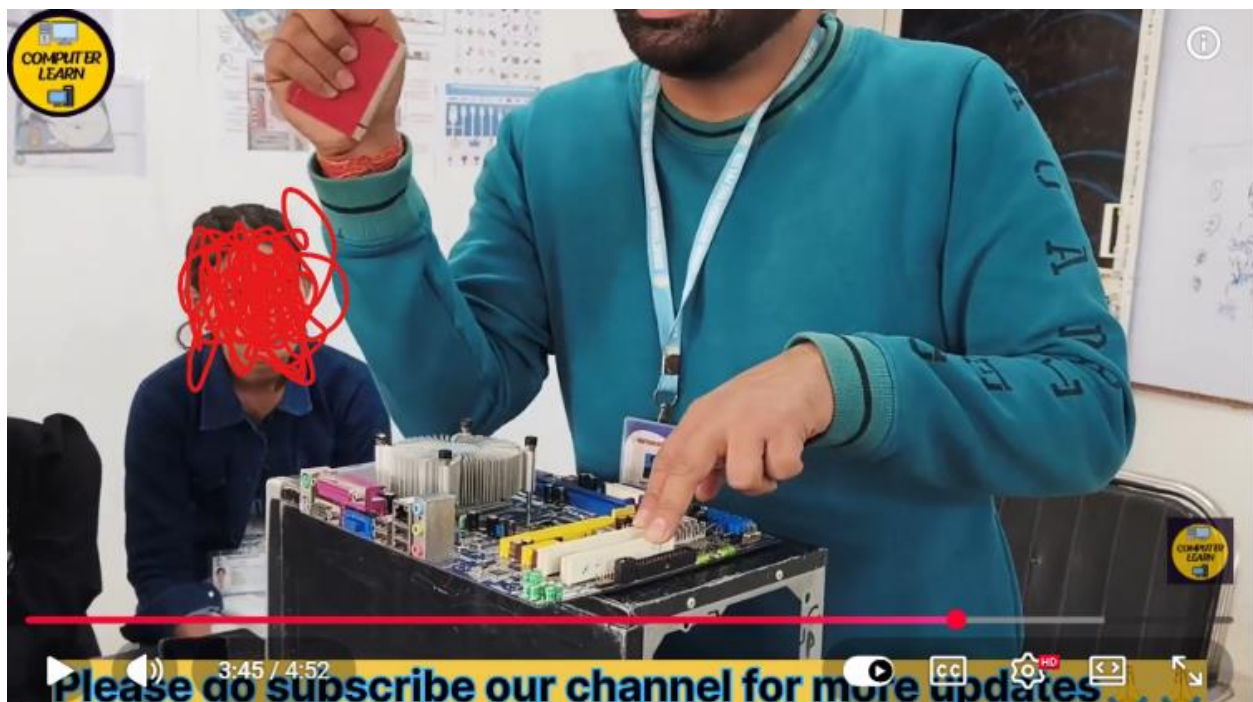


Figure 1: POST card demonstration showing diagnostic codes during motherboard startup.

Task 3: Software Diagnostic Tool

HWMonitor				
File View Tools Help				
Sensor	Value	Min	Max	
DESKTOP-C2E2V5Q				
Mainboard				
Temperatures				
TZT201	30.0 °C	30.0 °C	30.0 °C	
TZT200	28.0 °C	28.0 °C	28.0 °C	
Fans				
Fan 0	0 RPM	0 RPM	0 RPM	
Utilization				
Physical Memory Load	77.0 %	77.0 %	78.0 %	
Available Physical Me...	1.8 GB	1.7 GB	1.8 GB	
Virtual Memory Load	71.5 %	71.2 %	71.5 %	
Available Virtual Mem...	3.2 GB	3.2 GB	3.2 GB	
Intel Core i3 2130				
Voltages				
VDD (Max)	1.196 V	1.171 V	1.201 V	
Temperatures				
Cores (Max)	56.0 °C	51.0 °C	59.0 °C	
Package	56.0 °C	51.0 °C	59.0 °C	
Powers				
Package	14.99 W	9.11 W	21.82 W	
IA Cores	11.09 W	5.36 W	17.57 W	
GT	0.19 W	0.16 W	1.90 W	
Uncore	3.71 W	3.54 W	3.97 W	
Clocks				
Cores (Max)	3392.0 MHz	3392.0 MHz	3392.4 MHz	
CPU BCLK	99.8 MHz	99.8 MHz	99.8 MHz	
Memory	665.1 MHz	665.1 MHz	665.1 MHz	
Utilization				
Processor	28.5 %	10.1 %	41.4 %	
ST4000VX007-2D1166 (ZGY1Z8WL)				
Temperatures				
Assembly	34.0 °C	34.0 °C	34.0 °C	
Air Flow	34.0 °C	34.0 °C	34.0 °C	
Counters				
Power-On-Hours	20249 hrs	20249 hrs	20249 hrs	
Power Cycle Count	3026	3026	3026	
Utilization				
Used Space (d)	37.1 %	37.1 %	37.1 %	
Available Capacity (d)	629.0 GB	629.0 GB	629.0 GB	
Used Space (e)	22.1 %	22.1 %	22.1 %	
Available Capacity (e)	779.2 GB	779.2 GB	779.2 GB	
Used Space (f)	44.8 %	44.8 %	44.8 %	
Available Capacity (f)	400.2 GB	400.2 GB	400.2 GB	
Used Space (g)	0.3 %	0.3 %	0.3 %	
Available Capacity (g)	498.4 GB	498.4 GB	498.4 GB	
Used Space (h)	3.2 %	3.2 %	3.2 %	
Available Capacity (h)	484.0 GB	484.0 GB	484.0 GB	
Activity	0.0 %	0.0 %	0.1 %	
Speed				
Read Rate	0.00 MB/s	0.00 MB/s	0.00 MB/s	
Write Rate	0.00 MB/s	0.00 MB/s	0.00 MB/s	
FLEIN256GA87ECO (0523345028431)				
Temperatures				
Assembly	41.0 °C	41.0 °C	41.0 °C	
Sensor 1	54.0 °C	54.0 °C	56.0 °C	
Sensor 2	41.0 °C	41.0 °C	41.0 °C	
Counters				
Power-On-Hours	231 hrs	231 hrs	231 hrs	
Power Cycle Count	1041	1041	1041	
Health Status	98 %	98 %	98 %	
Utilization				
Used Space (c)	63.6 %	63.6 %	63.6 %	
Available Capacity (c)	86.1 GB	86.1 GB	86.1 GB	
Activity	0.7 %	0.0 %	3.3 %	
Speed				
Read Rate	0.00 MB/s	0.00 MB/s	0.48 MB/s	
Write Rate	0.02 MB/s	0.00 MB/s	2.98 MB/s	
Intel(R) HD Graphics				
Temperatures				
GPU	53.0 °C	50.0 °C	58.0 °C	
Clocks				
Graphics	848.0 MHz	848.0 MHz	997.6 MHz	
Utilization				
GPU	0.0 %	0.0 %	21.3 %	
Memory	76.9 %	76.9 %	76.9 %	
Video Engine	0.0 %	0.0 %	0.0 %	
3D	0.0 %	0.0 %	0.0 %	
Copy	0.0 %	0.0 %	0.0 %	
Realtek 8723DU Wireless LAN 802.11n USB NIC				
Speed				
Download Speed	7.91 Mbps	0.00 Mbps	20.82 Mbps	
Upload Speed	0.12 Mbps	0.00 Mbps	1.26 Mbps	
Realtek RTL8188FTV Wireless LAN 802.11n USB 2.0 Network Adapter				
Speed				
Download Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps	
Upload Speed	0.00 Mbps	0.00 Mbps	0.00 Mbps	

Description: I used HWMonitor to check the health information of my computer. It shows details such as CPU temperature, voltage, fan speed, and other hardware information.

Task 4: Comparison of Troubleshooting Tools

Tool	What it checks	Example Problem Detected
Multimeter	Checks electrical values such as voltage, resistance, and continuity.	A faulty power supply or broken cable.
POST Card	Checks the motherboard startup/POST process and displays diagnostic codes.	RAM, CPU or motherboard startup problems.
Software Diagnostics	Checks system health information such as temperatures, voltages and hardware status.	High CPU temperature or a storage health problem.

conclusion:

Each troubleshooting tool is useful for a different type of problem. A multimeter is mainly used for electrical testing; a POST card helps identify hardware problems during startup, and software diagnostic tools help monitor the health of the computer.